Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/24/2006
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Project Title:	Ultra-Slow and Stopped Light in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
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We have succeeded in slowing light pluses in a Bose-Einstein condensate (BEC) to only 17 m/s, more than seven orders of magnitude lower than the light speed in a vacuum. More recently, we completely stopped, storred, and retreived light pluses after storage times of several mimmiseconds. Associated with light speed reduction is an extreme spatial compression of pluses. In particular, with the most recent extension of our method, the 'light roadblock,' we have compressed light pluses from a length of 2 km to only 2ùm at the center of a condensate. At these length scales we can create extremely localized defects and generate dramatic excitations in the condensates. During the funding period we plan to perform ground-based experiments to pursue the possibilities for controlled storage and coherent processing of optical information in trapped BECs. We envision injection into a condensate of a time train of light pluses, in a two-dimensional pattern, with subsequent storage of the pluses to form highly compressed three-dimensional patterns of optical information. We plan to use atom-atom interactions for processing the information during Storage time, in

Task Description:	particular, we plan to control the interactions and processing with use of external electrical and magnetic fields. The light pluses will then be read out and the result of the processing deleted from (possibly dramatic) shape changes of the revived light pluses. The lowest light speeds--and thereby maximum spatial compression factors--obtainable in ground based experiments are limited by inhomogeneous effects due to the velocity spread in the atom clouds from finite temperature effects, and to Zeeman shifts from the magnetic trapping potential needed to keep the atoms in place. The maximum storage and processing times are also limited by finite temperature effects from incoherent smearing of the stored information and from dephasing and damping dure to the presence of non-condensed atoms surrounding the condensates. In a microgravity environment, we could cool atom clouds to much lower temperatures and could subsequently soften the magnetic trapping potential significantly, without causing sagging of the cloud due to gravity. We could produce condensates with macroscopic coherence lenghts, that could be trapped in the absence of non-condensed atoms. We could even turn off the trapping potential which would result in a very controlled atom cloud expansion, without a rapid acceleration and displacement of the cloud in gravity. For a carefully picked delay time after trap release, we could inject light pluses into a condensate of the exactly right spatial shape, a fact of extreme importance for studies of many-body systems of quantized vortices insuperfluid BECs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In the past year we have made significant progress on our NASA project. In particular, we have extended our slow light techniques to generate and observe a completely new class of non-linear superfluid excitations in sodium Bose-Einstein condensates (see publication list above). Our apparatus has been specifically developed to implement a double light roadblock to induce collisions between quantum shock waves generated at each of the two light roadblocks. This has resulted in the first direct observation of dark soliton collisions, quantized vortex ring collisions, soliton vortex-ring collisions, and the evolution of hybrid structures comprising both soliton segments and vortex rings [Ref.1]. This research will help physicists gain new insights into the superfluid phenomenon and into the breakdown of superconductivity. We plan to continue these studies of the rich superfluid dynamics of BECs. The out-of-equilibrium dynamics induced in the BECs result in the complex, compound collisional products that are dynamically created and are rather stable, hybrid structures made of solitons and multiple vortex rings (see Figure; to appear in Optics and Photonics News, May issue). The nonlinearity parameter of the medium, primarily determined by the density of the condensates, determines the stable nonlinear excitations in the system. Inspired by our work, our collaborators (J. Brand, S. Komineas, and N. Papanicolaou) have just recently worked out some of the theoretical dispersion curves for the lowest-energy, nonlinear excitations. These dispersion curves plotted as a function of the nonlinearity parameter show several branching points and competition between different types of hybrid soliton-vortex structures. We plan to apply the double light roadblock setup, and implement a high-resolution imaging system to image the dynamics of condensates in-situ in order to maintain better control of their density, with the option of imaging their dynamical sequences non-destructively. We can vary the experimental parameters to cover large ranges of the nonlinearity parameter and should be able to experimentally probe the dispersion curves of nonlinear excitations in condensates. We plan to go further and study collisions between the hybrid structures and hope to observe a whole hierarchy of increasingly complex collisional products. In the nonlinear media we are dealing with, interaction of nonlinear excitations are beyond the theory underlying the dispersion curves for single nonlinear excitations, and the study of these higher order collisional products are beyond current theoretical capabilities. The planned experimental collision studies could thus foster an extremely fruitful synergy between experiment and theory. The theoretical descriptions so far are based upon mean-field theory, and a particularly interesting question is: at which point in the hierarchy of collisional excitations would mean-field theory potentially break down. This also leads to the very important – and totally unanswered - question regarding the role played by quantum fluctuations in damping of the non-linear excitations. A micro-gravity environment would be of great importance because of the unique possibility it offers for eliminating disturbing effects from damping due to non-condensed atoms. Ref. 1 Observation of Hybrid Soliton Vortex-Ring Structures in Bose-Einstein Condensates. Naomi S. Ginsberg, Joachim Brand, and Lene Vestergaard Hau. Physical Review Letters, 94, 040403 (2005).

Bibliography Type:	Description: (Last Updated: 12/20/2019)
Articles in Peer-reviewed Journals	Ginsberg NS, Brand J, Hau LV. "Observation of hybrid soliton vortex-ring structures in Bose-Einstein condensates." Phys Rev Lett. 2005 Feb 4;94(4):040403. Epub 2005 Jan 31. PMID: 15783535 , Feb-2005
Articles in Peer-reviewed Journals	Ristroph T, Goodsell A, Golovchenko JA, Hau LV. "Detection and quantized conductance of neutral atoms near a charged carbon nanotube." Phys Rev Lett. 2005 Feb 18;94(6):066102. Epub 2005 Feb 17. PMID: 15783752 , Feb-2005
Articles in Peer-reviewed Journals	Dutton Z, Ginsberg N, Slowe C, Hau LV. "The art of taming light : ultra-slow and stopped light." Europhysics News. 2004 Mar-Apr;34:33-9. https://doi.org/10.1051/epn:2004201 , Mar-2004
Articles in Peer-reviewed Journals	Slowe C, Vernac L, Hau LV. "A High Flux Source of Cold Rubidium." Submitted to Rev. Sci. Instrum., March 2005. , Mar-2005
Articles in Peer-reviewed Journals	Dutton Z, Hau LV. "Storing and processing optical information with ultraslow light in Bose-Einstein condensates." Physical Review A 2004 Nov;70(5):053831. http://dx.doi.org/10.1103/PhysRevA.70.053831 , Nov-2004
Articles in Peer-reviewed Journals	Soljacic M, Lidorikis E, Hau LV, Joannopoulos JD. "Enhancement of microcavity lifetimes using highly dispersive materials." Phys Rev E Stat Nonlin Soft Matter Phys. 2005 Feb;71(2 Pt 2):026602. Epub 2005 Feb 8. PMID: 15783436 , Feb-2005
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Journal/Magazine covers	Dutton Z, Ginsberg N, Slowe C, Hau LV. "Cover in journal Europhysics News for article, “The art of taming light : ultra-slow and stopped light." Euophysics News 2004 Mar-Apr;34:33-9. , Mar-2004